pH and the pH Scale

pH, pOH and calculations for strong acids and bases

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS (25 °C)

$\mathrm{pH} = -\log[\mathrm{H_3O^+}]$ $\mathrm{pOH} = -\log[\mathrm{OH^-}]$ $\mathrm{pH} + \mathrm{pOH} = 14.00$ $[\mathrm{H_3O^+}] = 10^{-\mathrm{pH}}$

Give pH values to two decimal places.

A. Understanding

1. At 25 °C, classify each pH as acidic, neutral or basic: 2.5, 7.0, 9.8, 6.9.

2. How many times greater is $[\mathrm{H_3O^+}]$ in a solution of pH 2 than in a solution of pH 5?

B. Strong acids

3. Calculate the pH of 0.0010 M HCl.

4. Calculate the pH of 0.25 M $\mathrm{HNO_3}$.

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C. Strong bases

5. Calculate the pH of 0.010 M NaOH.

6. Calculate the pH of 0.0050 M barium hydroxide, $Ba(OH)_2$.

D. Concentration from pH

7. Calculate $[H_3O^+]$ for a solution with (a) pH 3.20 and (b) pH 11.50.

8. What is $[OH^-]$ in a solution with pH 5.00?

E. Challenge

9. 0.10 M HCl is diluted 100 times with water. What is the new pH? By how much has the pH changed?

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